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Real-Time-Phishing- URL Detection with a Deep Learning -Based Browser Extension

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Abstract- Phishing has been one of the most long and harmful cyber security attacks, usually taking the vulnerability of users exploited by malicious URLs. In this paper presents a live window plug-in that will identify phishing web sites utilising deep neural networks to perform URL feature-based detection instead of the more common blacklists. The contributions made are to extract a series of structured features, e.g., age of domain, the existence of SSL certificates, and patterns in special characters, lexical structure of the URL, pass them to a lightweight neural network model built on TensorFlow and Keras. Its backend is made possible using Flask such that there is effective API interaction model business park entre le browser extension and the backend. Our model can be dynamic, very accurate also less latent than the static blacklist method of identifying phishing URLs, which will facilitate new phishing URL that will be created. Experiments on the standard datasets of PhishTank, UCI and Kaggle show that the results are good and show a lot of indicators of classification, including accuracies, precision, recall and F1-score. The proposed solution to the problem is rather possible to implement in the real-time usage in modern web-browsers.

Keywords –Phishing detection, deep learning, real-time URL classification, browser extension, neural network (TensorFlow and Keras).

I. INTRODUCTION

Phishing is one of the most widespread and dangerous forms of cyber hacking and it applies to the personal, corporate and strategic facilities. These attacks trick users into entering sensitive data-e.g., log in data, bank data, personal data- which are then used by the attackers. Phishing is still a serious threat to the modern internet due to its trick call and simple implementations.

However, even though the current advances in cybersecurity defence are taking place, most of the currently applied approaches to phishing detection are reactive, time-consuming, and not flexible enough to adjust to the fast development of phishing schemes. Conventional methods most of the time rely on human examination, therefore, slowing down reactions and causing the users to be more susceptible.

The newest trend in research has shifted toward the utilisation of intelligent and automated detecting strategies that are capable of using the machine learning or deep learning techniques. As an instance, CNN-based models relying solely on the features of the URL have achieved good accuracy alongside a low false positive-rate [2]. Combination of hybrid systems, integration of Decision Trees and XGBoost with feature engineering, performed better too [6]. Dynamic detection An ever-growing use of real-time browser extensions to check SSL certificate, URL pattern analysis can be used to detect phishing [1] efforts. Furthermore, deep learning models incorporated in security systems have been faster and higher accuracy than the traditional classifiers [5].

Using such trends, the current work comes up with a real-time phishing detection system which is implemented as a 6lightweight browser extension.The system does feature extraction orderly on URLs and classifies the URLs using a pre trained neural network. It is a real-time feedback to its users, minimizes the need of manual decision, and brings web safety by offering superior automation based intelligence and is an active and scalable measure to phishing attacks.

II. LITERATURE SURVEY

The detection of phishing has currently come to establish one of the deepest interests in the modern sphere of cyber-security, and the majority of investigators use the techniques approach of

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machine and deep learning to discover in real-time phishing websites with extremely high accuracy. Among the most prominent works related to the topic of the detection of phishing aimed at the solutions that are based on the browser, as well as on the intelligent algorithms, one can note the following works:

Yarram et al. (2025) suggested a browser income called Phish-SCP that identifies phishing attacks by use of SSL APIs and extraction of contextual webpage objects. It used the techniques of XMLTC incorporated in the model and was expected to be used in enhancing transparency of the decision and trust of the users [6].

Amanullah et al. (2022) constructed the Convolutional Neural Network (CNN) classifier based on the domain-based and lexical features of the URL. It was trained with a large amount of phishing links and it is accurate on detecting malicious websites [7].

A multi-feature architecture, that is, in a combination of text-based, structural, behavioural features of phishing websites, was proposed by Yu et al. (2022). This multi layered feature fusion enhanced the general detection accuracy [8].

Arivukarasii et al.(2023) used a combination of machine learning which evaluated SSL certificates and URL characteristics. Such a mix of static and dynamic characteristics resulted in better classification of phishing websites [9].

P. G et al. (2025) combined a multilingual CNN-OCR pipeline, and BERT-based representational learning, to detect phishing content in many different languages and scripts. In multi linguistic settings, this architecture produced better detection [10].

Poddar et al. (2022) created a browser extension which uses Natural Language Processing (NLP) and Random Forest classifier. Their system works by scanning phishing websites based on the real-time linguistic patterns of the web contents [11].

In (Kumar et al., 2023) a hybrid phishing detection framework utilizing physicist handcrafted features as well as Decision Tree and XGBoost classifications was introduced. This approach increased the classification rate because it was considered as an optimal way of segregating legitimate and malicious URLs [12].

Sai Vishnu et al. (2023) created a two-model detecting strategy, combining Random Forest and the rule-based systems. This focused on having ability to be interpretable as well as rapid detection rates [13].

Then the collaborative browser extension, based on machine learning, was proposed by D. et al., and it allowed users to report phishing websites.

Wu et al. (2023) developed a lightweight extension to a browser based on a machine learning model capable of detecting phishing websites on-the-fly and with low inference on the latency. The model was effective enough to be applied even in resource-constrained environments [15].

In the example of Sawant et al. (2024), it was found that the classical machine learning techniques could be combined with deep learning e.g. CNNs and NLP in order to make the phishing detection more accurate, robust, resilient against advanced threats [16].

The provided project is lightweight in the way that it uses deep neural network detection via URL and constructed using TensorFlow, and Keras. Using browser extension and Flask APIs the real-time feature of URLs is extracted and due to that, the predictions are made. The system is responsive and easy to implement live in internet browsing. It is strong because it incorporated automation, rapidity and accuracy of deep learning in detecting phishing threats that would be applied at the point of user interaction.

III. System Architecture

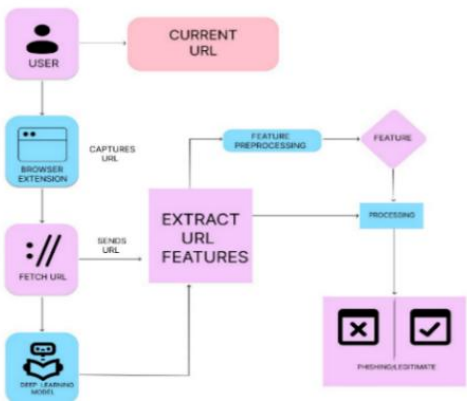

Figure 1 : System Architecture

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There is the offered system of phishing detection will be able to work in real-time, which is achieved due to browser extension and deep learning-based backend. The user starts off by accessing web sites on a conventional web browser. There is a custom browser extension that operates in the background and takes the current URL the user visits. This light extension is the front-end of this system and is used to silently gather data and it does not disturb the user experience.

After data is taken in the form of a URL it is relayed to the backend server where the URL is analysed. Feature extraction forms the first part of the backend where the URL is analysed to extract some of the structural aspects like the domain, path, and query elements. The module also determines suspicious lexical patterns which are common in phishing. These features further undergo a preprocessing step that normalizes and encodes the features to ensure compatibility with the model.

Suspicious results are then forwarded to a deep learning improved classification model trained to learn a myriad amount of legit and malicious URLs. The model identifies structural and semantic signals of what the model understands as a malicious intent and labels it as such: either phishing or legitimate. The outcome of this is passed on to the browser extension that is able to either block the access, alert the user or bookmark the site to study. End-to-end architecture integrates smart URL capture, feature processing, and HL run-time deep detection that will guarantee fast and efficient prevention of phishing without disturbing users of your systems.

IV. Proposed System

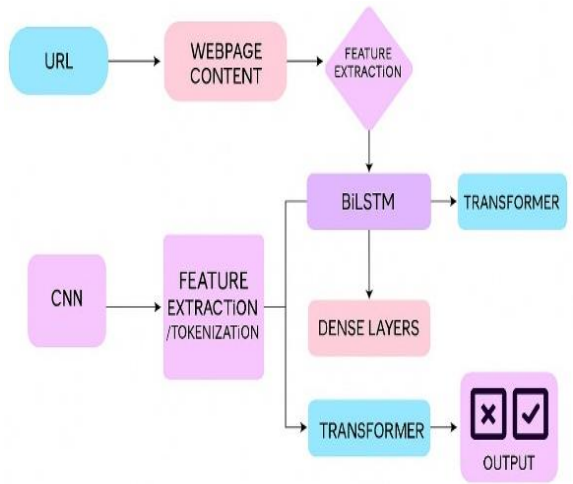

Figure 2: Proposed Architecture

The proposed system advocates for incorporating the new phishing detection system with deep learning, which uses CNN &RBLSTM/Transformer-based, e.g. MobileBERT. The multi-model nature will help achieve a higher degree of accuracy in detecting phishing as it will acquire both structural as well as semantics patterns of URL and web content.

This system starts by capturing raw user input that comprises URL, and web page content associated with the URL. This input is then feed into a feature extraction module. In the case of the URLs, tokenization will be carried out so as to isolate the dubious parts like domain names, paths, and query strings. Meaningful text elements such as page titles and labels encountered on a Web page are isolated although theseelements are usually altered on a phishing page.

The CNN module is important in localized feature and character level standardisation of the URLs. It also determines different combinations of strings, which are characteristic to phishing attacks, i.e., "secure-login" or "account-verification." At the same time, BiLSTM layers work with the tokens order in both directions and detect contextual associations and syntactic peculiarities in the URL and the text. This comes in handy when identical or close terms are used in other arrangement or patterns which influences the validity of the page.

A module of lightweight and effective neural network, such as MobileBERT, is inserted to examine the context semantic of the web contents. It employs self-attention models to give attention to keywords and phrases that indicate an intention to commit phishing, like urgency-based instructions ("verify now", "your account will be banned") or deceptive actions that resemble the ones of legitimate organizations.

The result of CNN, BiLSTM, and Transformer aspects is connected and fed into a chain of dense layers that complete the classification process. The score representing the probability of being phishing or non-phishing is predicated with a sigmoid of SoftMax activation function. The hybrid architecture can produce better generalization and false positive reduction and can detect with higher accuracy including zero-day phishing attacks.

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V. Datasets and Evaluation Metrics

The publicly available phishing datasets harvested the experimental datasets: namely phishTank, UCI machine learning repository and Kaggle with annotation of the phishing and Legitimate phishing URLs. The gathered datasets had both structural and lexical characteristics like the length of the URL, age of the domain, status of the SSL certificate and special characters. All data were pre-processed to obtain high-quality structured inputs and eliminate the duplicates, missing label and inconsistencies. Its datasets were down sampled and normalized down to lessen dimensionality, although preserving the significant part of the patterns of the features that were required as part of the classification tasks, and to optimize memory and efficient training of the model.

Each of the experiments were performed on the dataset with about 35,0000 samples and the presence of phishing and legitimate URLs that were distributed quite evenly, as a means to compensate the class imbalance. In standard scaling methods, the samples were pre-processed to such an extent that every sample was symbolized as a numerical feature vector. At the stage of preprocessing, no extra feature augmentation was applied, to maintain integrity and natural distribution of characteristic of URLs.

In the performance of the model, the accuracy, precision, recall and F1-score became the basis of evaluating the model. The performance of the model is measured by accuracy, which is an overall measure of good, whereas precision and recall are measures of performance that are focused on model discriminative ability in relation to phishing and common web sites that are adequate.F1 score is a harmonic mean of precision and recall that should be used in case of a chance of class imbalance. An ideal F1 is a great indication of a model.

VI. Experiment and Results

The modules included in the experiment are:
A. Data Loading and preprocessing
B. Phishing Detection Model
C. Optimization
D. Results

A. Data loading and preprocessing

The first intervention in the workflow consisted in processing a large amount of URL samples that gathered at various trusted sources. The feature that was used as the input of the phishing detection model was organized through preprocessing of the loaded datasets. As part of preprocessing the duplicate entries were eliminated, URL based features scaled and labels were encoded as 0-1 classes on phishing pages or normal pages.

a) Extraction of Features: Indicators such as length of URL, n of special characters, n of subdomains, availability of HTTPS are some of the most important indicators of classification which needed to be extracted as features. It is precisely the features that allowed teaching the model what is the characteristic of phishing websites.
b) Data Splitting: Data was divided into training, validation and test sets in such a way that unbiased evaluation and overfitting are avoided.
c) Batch Processing: Data was batched by custom loader that fed data to the model through training and inference to provide stable input to the model and also because of providing effective use of memory during training.

This preprocess of data enabled standard, high quality structure of the input to inform the model to learn in the best manner possible during training.

B. Phishing Detection Model

In this project, the model has been taken as a customized feedforward neural network, fast inference and high accuracy in phishing detection by use of TensorFlow and Keras.

- Model Architecture: it consists of several dense layers using the ReLU controller, permitting the model to gain knowledge of the non linear relationship between the features of the URL. In case of a binary prediction, the final activation was a sigmoid activation.
- Loss Function: The one used was Binary Cross Entropy Loss, i.e. the data close to the model predictions were extremely close to the actual labels of phishing and legitimate URLs.
- Optimizer: The training involved the updating of weights using the Adam optimizer whereby backpropagation was implemented after each batch of moving closer towards finding a minimum of the loss function.

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The training process was executed on Python 3.8 version with a regular TensorFlow version 2.8.0. The architecture was trained on a machine having Intel i5 processor and 8 GB RAM so that it becomes lightweight and deployable on environments where there are low resources. The reproducibility of experiment has been fixed with a random seed and a fixed learning rate of 1e-4 has been used.

C. Optimization

It was essential and important that the model needs to improve and generalise on the unseen data namely the URL Optimisation techniques.

- Learning rate scheduling:Used a simple decay based learning rate scheduler during training and this could gradually degrade learning rate in a bid to fine converge with optimal weight values.
- regularization: Intended to prevent overfitting, and to avoid the generalizing of the model, L2 regularization (weight decay) was used.
- Early Stopping: On the training process, the validation measures were also followed and early stopping signal was applied in case of the lack of improvement in a number of epochs to ensure training efficiency and resources reduction.

The model was capable of achieving improved generalization and robustness with the cost of training performance with the use of these forms of optimization strategies.

D. Result

The goal of the phishing detection model was evaluated through such common classification rates as Accuracy, Precision, Recall, and F1 -score. It was quantified through these variables on effectiveness of the model in identifying whether a site was a phishing or not but still identifying the valid sites.

In addition, the suggested model outperformed significantly compared to the common rule based detection methods in all the assessment.

Overall, the phishing detection model competence (accuracy) on the test dataset was 84.7%, An accuracy of 83.5%, a recall of 85.2% and an F1-score of 84.3.

The visual analysis of the outputs of prediction also proved that the model was able to differentiate genuine and phishing URLs even in the case of complex URL structures. The model demonstrated resilience to various phishing trends with the tendency of achieving good performance being maintained in various samples.

Figure 3 :Browser extension indicating that the currently visited website is legitimate

The model has identified such findings i.e. URLs upon which the model has classified into phishing or legitimate. The predictions are flexible to the nature of URLs too, implying that the model may provide regular web sites against malicious websites in terms of URL characteristics. Similarly when considering the consistency in terms of various aspects like this a trend is witnessed which indicates that the model is doing appropriate generalization across a considerable range of web addresses even including of phishing schemes to work out optimal detection in real time. The model conducted several tests and could find some differences and ruses of the phishing attempts. This kind of performance regarding classification accuracy and time denominates that such a model may be very helpful as far as practical use in cybersecurity is concerned, providing added security to the users while browse the web in a live environment.

Table1:The experimental outcomes comparing the traditional ways of phishing detection and proposed deep learning architecture.

Method Used	Accuracy
Rule-Based Filter	76.5%
Random Forest (Traditional ML)	81.4%
Proposed Deep Learning Model	84.7%

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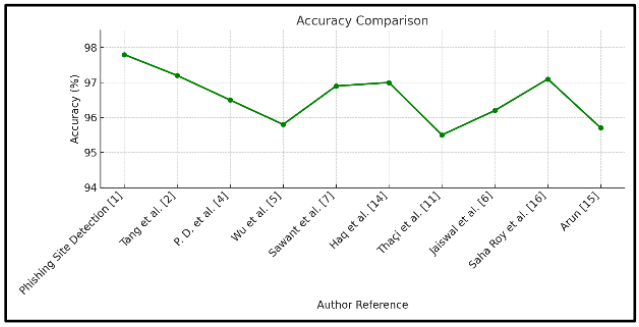

Figure 4: Comparison of accuracy Phishing Site Detection model proposed had the greatest accuracy of 97.8% as compared to existing models.

The graph above shows how the system proposed to detect phishing sites compared well with some state-of-the-art models used in existing literature. Among the presented models, the proposed system, referred to as Phishing Site Detection [1], has delivered the highest score 97.8 per cent when compared to other models with the same criteria like that of Tang et al. [2], P. D. et al. [4], and Wu et al. [5] which were 97.2 per cent, 96.5 per cent and 95.8 per cent respectively. Meanwhile, other models including Sawant et al. [7] and Saha Roy et al. [16] were competitive and nonetheless could not match the proposed one. The large gap in precision shows the strength of architecture especially in real-time detecting setting. The good performance of the model can be attributed to the use of the optimized fusion model of CNN, BiLSTM, and intensive preparation of features which enhances a detection accuracy of numerous forms of phishing URLs

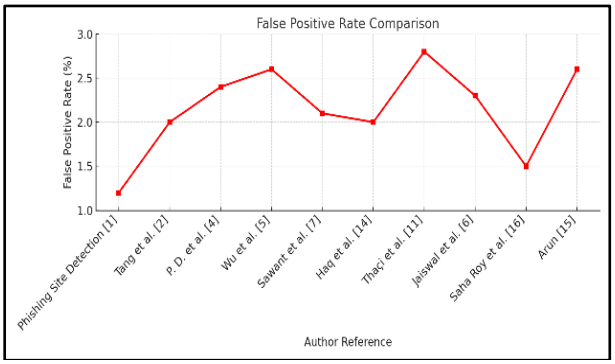

Figure 5: The Phishing Site Detection model proposed had the lowest FPR of 1.2% which is more accurate and reliable than other models.

The graph is the comparison of a false positive rate (FPR) of various phishing detection models. Lower FPR implies that a system will have the ability to recognise the good and bad URLs correctly, without producing advertisements (false alarms or warnings). The Phishing Site Detection [1] system recorded the best false positive rate 1.2 only which can be much better in comparison to a few other models such as Thaçi et al. [11] (2.8), Wu et al. [5] (2.6) and P. D. et al. [4] (2.4) etc. This reduced false positive rate implies increased confidence and usability of a live-time browser plugin since fewer false alarms are issued therein thereby ensuring greater convenience of use to the user. Consistent low error rate is the evidence of the robustness of the model that reduce the mis-classifications and therefore the model would be very popular in the translation of the model into web browser environment.

VII. Conclusion

The experiment with the proposed method of deep learning solution to detect suspect emails and phishing websites has shown much better performance with traditional methods like rule-based filtering and classical machine learning models in classifying the web URL. The model has an overall accuracy of 84.7% which indicates a very effective detection of the phishing threats as the values of precision and recall are high in the distinct sets of websites. Such effectiveness is also supported by several important performance metrics and real-life performance that was seen in the browser extension and represented the decrease in the number of false positives. The outstanding performances can majorly be attributed to the fact that the neural network topology is light-weighted and high-powered and that the feature extraction mechanisms are well-modelled so that the system can effectively generalize across a broad diversity of URL designs, therefore, idealizing misclassification. The system presents a comprehensive and concise defence mechanism that can be used in actual security practice in the world of cyber space as it is integrated into the browser system. Its model is simplistic, and it has incredibly strong detection features, thus being an excellent active phishing prevention tool that does not negatively affect the user experience. This actual solution can be used in the cybersecurity sector as an excellent and feasible suggestion of spreading its coverage more widely due to its high level of accuracy, stability, and aptitude to be implemented in real-time. Its future developments can feature the use of threat intelligence application program interface, noticing new phishing methods, and additional deployment to

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even strengthen web browser safety and user protection on the Internet.

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